False Confidence in Generative Samplers: Exact Physical References Expose Hidden Errors and Diagnostic Blindness

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Abstract

Learned samplers are increasingly proposed for equilibrium sampling and free energies in chemistry, and they are usually judged by self-normalized diagnostics such as the effective sample size (ESS). We show with an elementary proposition that these diagnostics are structurally invariant to the Boltzmann mass α of unvisited basins: conditioned on not proposing a basin, the importance weights follow the law of the truncated system, so the ESS stays high while the free energy shifts by $-k_BT\log(1-\alpha)$ and the nominal delta-method interval contracts around the biased value, yielding asymptotically zero coverage. In a toy double well where one basin holds 49.9% of the mass, a proposal putting mass 10^{-6} on that basin achieves a median ESS of 98% but covers the exact $\log Z$ in 0% of 400 runs. On exactly enumerated Sherrington-Kirkpatrick instances ($n = 24$), nominal 95% intervals from masked diffusion and a GFlowNet cover the truth in only 8% and 47% of runs at $\beta = 2$, while classical estimators and an autoregressive network (VAN-MADE) stay near 95%. A reverse check that scores the proposal on draws from an independent reference sampler flags the dropped basin where the forward Jensen gap is blind. We demonstrate this on a continuous molecular model: a RealNVP Boltzmann generator trained on the alanine dipeptide Ramachandran landscape drops the $C_{7\mathrm{ax}}$ basin (8.1% mass), reporting a median ESS of 79% with only 10% coverage, while the reverse check detects the omission ($G_{\mathrm{rev}} = 11.7$ nats). Finally, we size the bound on macromolecular systems (chignolin, Trp-cage, LJ_{38}), contrasting single-trajectory kinetic trapping with converged Anton and NMR equilibrium baselines to show that dropped macroscopic states shift free energies by 0.68 to 1.02 kcal/mol, exceeding lead-optimization thresholds.

1 Introduction

Normalizing flows, diffusion models and generative flow networks (GFlowNets) are proposed as samplers of Boltzmann distributions $p(x) \propto e^{-\beta U(x)}$ for molecules and materials (Noé et al., 2019; Hoogeboom et al., 2022; Satorras et al., 2021; Lipman et al., 2023). The partition function Z of such systems is unknown, so a learned sampler is usually judged by statistics of its own importance weights: the effective sample size (ESS), the Pareto $\hat{k}$, and delta-method intervals for the free energy.

These statistics share a critical failure mode: they are computed exclusively from configurations the sampler proposed, so a basin it never proposes leaves no trace in them. Section 3 formalizes this diagnostic blindness: conditioned on never visiting a basin of Boltzmann mass α, the importance weights follow the law of the problem with that basin deleted. As a result, the free-energy estimate is shifted by $-\beta^{-1}\log(1-\alpha)$ while the ESS remains high and nominal confidence intervals shrink at rate $O(M^{-1/2})$, guaranteeing asymptotic 0% coverage around the wrong value (Figure 1). While support coverage is an elementary requirement of importance sampling, modern evaluation culture has overlooked a critical consequence: self-reported diagnostics falsely reward mode collapse with high confidence, reporting near-optimal ESS and contracting intervals precisely when macroscopic states are omitted.

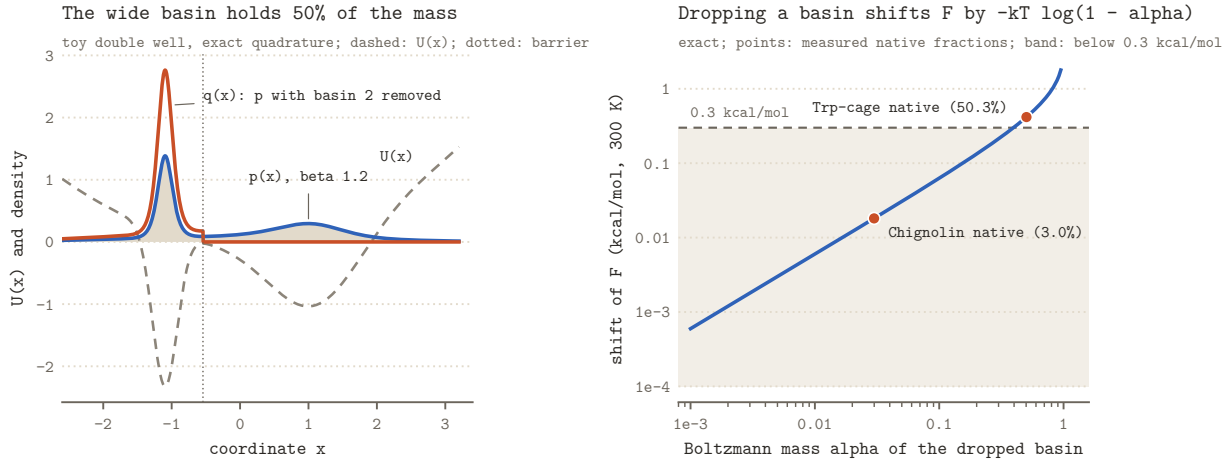


Figure 1: Left: the toy double well of Section 6 at $\beta = 1.2$. The narrow basin is the energy minimum, but the wide basin holds 49.9% of the Boltzmann mass (exact quadrature). Shaded: p. q is p with the wide basin removed. Right: shift of the free energy at 300 K when a basin of mass α is never proposed, $-k_B T \log(1 - \alpha)$ (exact). Points mark native-basin fractions; shaded: shifts below 0.3 kcal/mol.

To determine how this pathology manifests across physical systems, we organize our evaluation into three tiers:

1. **Exact empirical audit on disordered systems (Section 5):** Spin glasses feature frustrated, multi-basin landscapes where ground-truth partition functions can be enumerated exactly up to $n = 28$ spins. On these references, masked diffusion and GFlowNets exhibit severe calibration collapse, covering exact values in only 8% and 47% of runs at $\beta = 2$. In contrast, classical thermal estimators and normalized autoregressive networks (VAN-MADE) remain calibrated near 95%.

2. **Continuous neural molecular sampler (Section 6):** On the 2D Ramachandran landscape of alanine dipeptide, a RealNVP Boltzmann generator trained via reverse-KL completely drops the $C_{7\text{ax}}$ basin (8.1% mass), yet reports a median ESS of 79% while nominal interval coverage collapses to 10%. An independent reverse check immediately flags the omission ($G_{\text{rev}} = 11.7$ nats).

3. **Macromolecular landscape sizing (Section 7):** We map the theoretical bound to chemical units (kcal/mol) across all-atom protein landscapes (chignolin, Trp-cage) and the LJ_{38} cluster. We contrast single-trajectory kinetic trapping with converged Anton and NMR equilibrium baselines, proving that omitting macroscopic folded states shifts free energies by 0.68 to 1.02 kcal/mol, well above medicinal chemistry targets.

Contributions. (i) A formal proposition proving that self-normalized diagnostics and confidence intervals are blind to unvisited Boltzmann mass (Section 3). (ii) A combinatorial benchmark comparing learned versus classical samplers against exact ground truth (Section 5). (iii) A bidirectional audit protocol that unmasks dropped basins in toys and continuous molecular flows (Section 6). (iv) Biophysical scaling that anchors free-energy shifts to converged protein equilibrium benchmarks (Section 7).

2 Related work

Neural samplers. Variational autoregressive networks (Wu et al., 2019) and Boltzmann generators (Noé et al., 2019) introduced learned samplers for physical systems. Later work uses equivariant diffusion (Hoogeboom et al., 2022; Satorras et al., 2021), flow matching (Lipman et al., 2023; Albergo & Vanden-Eijnden, 2023; Bose et al., 2023), discrete diffusion (Austin et al., 2021; Zhu et al., 2025; Du et al., 2026), annealed transport (Holderrieth et al., 2025) and GFlowNets (Bengio et al., 2021; Malkin et al., 2022).

Free-energy estimators. Bennett's acceptance ratio (Bennett, 1976) and its multistate form (Shirts & Chodera, 2008) combine forward and reverse samples, and the Jarzynski and Crooks relations (Jarzynski, 1997; Crooks, 1999) connect nonequilibrium work to free energies. The reverse check of Section 6 uses the same two-sided idea. Xu (2026) proposes a ground-truth-free check for mode collapse, and Fu et al. (2023) shows that force accuracy does not guarantee correct simulations for learned force fields.

Spin glasses. The Sherrington-Kirkpatrick (SK) model (Sherrington & Kirkpatrick, 1975) has a frustrated landscape whose free energy in the limit $n \to \infty$ is given by the Parisi formula (Parisi, 1980), proven by Guerra (2003) and Talagrand (2006). For small n the partition function is computed exactly by enumeration.

3 Unvisited support

Let $(\Omega, \mathcal{F}, \mu)$ be a configuration space endowed with a dominating reference measure μ (Lebesgue measure for continuous coordinates, counting measure for discrete spins). Let $U : \Omega \to \mathbb{R}$ be a measurable potential energy and $p(x) = e^{-\beta U(x)}/Z$ the Boltzmann probability density with respect to μ, with partition function $Z = \int_\Omega e^{-\beta U(x)}\, d\mu(x)$ and free energy $F = -\beta^{-1} \log Z$. A learned proposal density q_θ (with $q_\theta(x) > 0$ for μ-almost all x where $p(x) > 0$) is trained to approximate p, for example by minimizing $\mathrm{KL}(q_\theta \,\|\, p)$ (Noé et al., 2019). Given i.i.d. draws $x_1, \ldots, x_M \sim q_\theta$, the importance-sampling estimate is

$$w_k = \frac{e^{-\beta U(x_k)}}{q_\theta(x_k)}, \qquad \hat{Z}_M = \frac{1}{M} \sum_{k=1}^{M} w_k, \qquad \hat{F}_M = -\beta^{-1} \log \hat{Z}_M, \tag{1}$$

its reliability is judged by the self-normalized effective sample size

$$\mathrm{ESS}_M = \frac{\left(\sum_k w_k\right)^2}{\sum_k w_k^2}, \tag{2}$$

and a nominal 95% interval is $\hat{F}_M \pm 1.96\, \beta^{-1} \hat{\sigma}_w / (\sqrt{M}\, \hat{Z}_M)$, where $\hat{\sigma}_w$ is the sample standard deviation of the weights.

Proposition 1 (Diagnostic blindness under unvisited support)**.** *Let $A, B \in \mathcal{F}$ form a measurable partition of Ω ($A \cap B = \emptyset$, $A \cup B = \Omega$) with $p(B) = \alpha \in (0, 1)$ and $q_\theta(B) = \epsilon \in [0, 1)$. Let $\mathcal{E}$ be the event that no draw lands in B (all M draws land in A). Then*

1. *$\mathbb{P}(\mathcal{E}) = (1 - \epsilon)^M$, which tends to 1 as $\epsilon \to 0$ for fixed M;*
2. *conditioned on $\mathcal{E}$, the draws are conditionally i.i.d. with density $q_A(x) = q_\theta(x)\mathbf{1}_A(x)/(1 - \epsilon)$;*
3. *conditioned on $\mathcal{E}$, $\mathbb{E}[\hat{Z}_M \mid \mathcal{E}] = (1 - \alpha)Z/(1 - \epsilon)$, and by continuous mapping $\hat{F}_M \xrightarrow{p} F - \beta^{-1} \log \frac{1-\alpha}{1-\epsilon}$ as $M \to \infty$, which approaches $F - \beta^{-1} \log(1 - \alpha)$ when $\epsilon \ll \alpha$;*
4. *conditioned on $\mathcal{E}$, the joint law of the normalized weights $\tilde{w}_k = w_k / \sum_j w_j$, and hence of ESS_M and of the relative interval width $\hat{\sigma}_w / \hat{Z}_M$, depends on p and q_θ solely through their conditional restrictions on A, and is strictly invariant to α and to the potential landscape on B, provided U on A is held fixed.*

Proof. Because the draws x_k are i.i.d., $\mathbb{P}(\mathcal{E}) = \prod_{k=1}^{M} \mathbb{P}(x_k \in A) = (1 - \epsilon)^M$. Conditioned on the product event $\mathcal{E}$, each draw has conditional density $q_\theta(x \mid x \in A) = q_A(x)$ independently. For $x \in A$,

$$\mathbb{E}_{q_A}[w] = \int_A \frac{e^{-\beta U(x)}}{q_\theta(x)} \frac{q_\theta(x)}{1 - \epsilon}\, d\mu(x) = \frac{Z_A}{1 - \epsilon}, \qquad Z_A = \int_A e^{-\beta U(x)} d\mu(x) = (1 - \alpha)Z. \tag{3}$$

By the weak law of large numbers conditioned on $\mathcal{E}$, $\hat{Z}_M \xrightarrow{p} (1 - \alpha)Z/(1 - \epsilon)$. Since $g(z) = -\beta^{-1} \log z$ is continuous on $(0, \infty)$, the continuous mapping theorem yields $\hat{F}_M \xrightarrow{p} F - \beta^{-1} \log \frac{1-\alpha}{1-\epsilon}$, proving item 3. For

item 4, write $e^{-\beta U(x)} = Z_A\, p(x \mid A)$ for $x \in A$. Then $w_k = c\, p(x_k \mid A)/q_A(x_k)$ where $c = Z_A/(1-\epsilon)$ is a positive scaling constant. The normalized weights $\tilde{w}_k = w_k / \sum_j w_j = \frac{p(x_k|A)/q_A(x_k)}{\sum_j p(x_j|A)/q_A(x_j)}$ cancel c identically, so their joint distribution depends on p and q_θ solely through their restrictions to A. Changing α or altering U on B while preserving $U|_A$ only shifts c, leaving the joint law of the normalized weights, ESS_M, and $\hat{\sigma}_w/\hat{Z}_M$ strictly unchanged. □

Diagnostic consequence: coverage collapse. The practical danger is not merely that importance sampling requires support coverage: it is the structural blindness of self-reported metrics. By item 4, if a model places negligible mass $\epsilon \ll 1/M$ on a basin, the event $\mathcal{E}$ occurs with high probability ($> 99.9\%$ for $\epsilon = 10^{-6}$, $M = 10^3$). Conditioned on $\mathcal{E}$, the sample variance $\hat{\sigma}_w^2$ and tail diagnostics such as Pareto $\hat{k}$ (Vehtari et al., 2024) reflect only the fit within A. In any regime where $\epsilon \ll 1/M$ as sample size M grows, the nominal delta-method confidence interval contracts at rate $O(M^{-1/2})$ around the biased target $F - \beta^{-1}\log(1-\alpha)$, while ESS_M/M can approach 100% and $\hat{k}$ remains falsely reassuring (< 0.5). Consequently, the empirical coverage of the nominal 95% interval collapses asymptotically to 0%.

In continuous state spaces, A and B are Borel measurable sets, and $q_A(x) = q_\theta(x)/\int_A q_\theta$ is well-defined whenever $q_\theta(x) > 0$ on A. While continuous neural proposals typically have non-zero support everywhere due to diffuse Gaussian tails, this pathology remains operant whenever a physical basin is separated by high free-energy barriers or entropic bottlenecks. If the proposal assigns probability $\epsilon \ll 1/M$ to that basin, no finite sample will visit it, leaving the empirical weights truncated and the diagnostics blind.

The right panel of Figure 1 illustrates the scale of this bias: at 300 K a dropped basin must hold about 50.3% of the mass (as Trp-cage does in our trajectory) to shift F by 0.417 kcal/mol, exceeding common chemical accuracy targets, while a basin holding 3.0% (chignolin in our trajectory) shifts F by only 0.018 kcal/mol. Related failure modes for mode-collapsed discrete samplers were observed by Xu (2026).

4 Why spin glasses are a fair test

A natural objection is that discrete Ising spins on a complete graph are physically distinct from continuous molecular conformations in three dimensions. The mathematical statement of Proposition 1 is coordinate-free: it requires only that the target distribution possess multiple metastable basins separated by barriers, and that the proposal fails to populate at least one of them. What an empirical benchmark requires is a system with many such frustrated basins where the true partition function is known exactly, without relying on approximate reference simulations.

The Sherrington-Kirkpatrick (SK) model, $H_n(\sigma) = n^{-1/2}\sum_{i<j} g_{ij}\sigma_i\sigma_j$ with $g_{ij} \sim \mathcal{N}(0,1)$, provides exactly this setting. Below the critical temperature ($\beta > 1$), its free-energy landscape fractures into a hierarchical ultrametric tree of pure states with extensive Hamming barriers (Parisi, 1980; Talagrand, 2006). For $n \leq 28$ spins ($2^{28} \approx 2.68 \times 10^8$ configurations), the complete state space can be enumerated on GPUs, yielding exact partition functions and Boltzmann weights.

We do not claim that mean-field spin glasses replicate the Euclidean geometry, bond constraints, or continuous transition paths of atomistic proteins. The role of the SK model is a rigorous falsification test: if an architecture or diagnostic cannot reliably detect the omission of macroscopic states in an exactly-solvable system with known truth, it provides no mathematical or empirical justification for confidence on continuous molecular landscapes where ground truth is unknown.

5 Exact audit on spin glasses

We evaluate the main test split of the SpinGlassBench benchmark at $n = 24$: Gaussian SK instances, seven inverse temperatures $\beta \in \{0.3, 0.6, 0.9, 1, 1.2, 1.5, 2\}$, an equal budget of 10^5 energy evaluations per estimate across all methods, exact $\log Z$ computed by GPU enumeration, and 5 seeds per instance. Classical estimators were evaluated across 200 instances (1,000 runs total); computationally intensive learned methods were trained and evaluated across 20 to 30 instances (100 to 150 runs total). Table 1 and Figure 2 report summary metrics from this benchmark.

Table 1: SK, $n = 24$, test split, 10^5 energy evaluations per estimate, 5 seeds per instance. Error: mean $|\log \hat{Z} - \log Z|/n$. Coverage: share of nominal 95% intervals containing exact $\log Z$ (Clopper-Pearson 95% binomial CIs in text); n/a: method reports no interval. Seconds: mean wall-clock per estimate at $\beta = 1.5$, one thread; learned methods include training.

Method	$\lvert\log \hat{Z} - \log Z\rvert/n$		Coverage (%)		Seconds
	$\beta = 0.6$	$\beta = 1.5$	$\beta = 0.6$	$\beta = 1.5$	per estimate
SA-TI	2.4×10^{-4}	7.1×10^{-4}	n/a	n/a	0.06
PT-TI	2.8×10^{-4}	8.7×10^{-4}	n/a	n/a	0.11
AIS	2.7×10^{-4}	8.9×10^{-4}	96	94	0.07
SMC	2.7×10^{-4}	8.8×10^{-4}	96	95	0.07
PA	2.7×10^{-4}	7.8×10^{-4}	96	96	0.08
VAN-MADE	2.7×10^{-5}	8.6×10^{-5}	95	96	2.49
Masked diffusion	1.6×10^{-4}	1.3×10^{-2}	94	60	6.96
GFlowNet	1.5×10^{-3}	2.4×10^{-2}	94	71	30.60

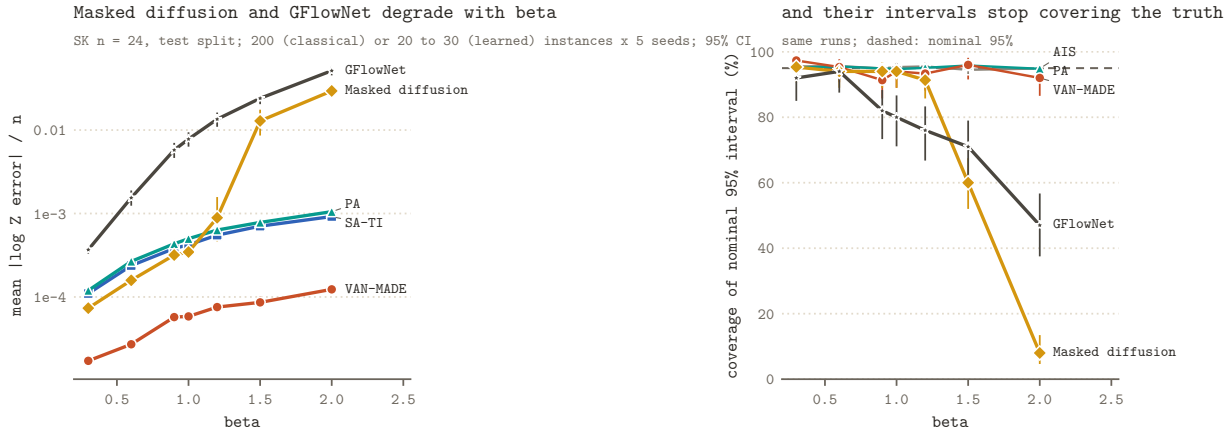


Figure 2: Left: mean absolute error of $\log Z$ per spin against β. Right: coverage of nominal 95% intervals. SK $n = 24$, test split, 5 seeds per instance; error bars are 95% intervals over instances. Dashed: nominal 95%.

Accuracy and architectural divergence. At $\beta = 1.5$, error per spin is 7.1×10^{-4} for simulated annealing with thermodynamic integration (SA-TI) and 7.8×10^{-4} for population annealing (PA). Crucially, learned samplers show sharp architectural divergence:

- **Normalized autoregressive networks (VAN-MADE):** Achieve the highest accuracy per energy evaluation (8.6×10^{-5}, roughly 8 times lower error than SA-TI) and maintain calibrated coverage across all temperatures: 95% at $\beta = 0.6$ (95% Clopper-Pearson CI: $[88.7\%, 98.4\%]$), 96% at $\beta = 1.5$ ($[90.1\%, 98.9\%]$), and 92% at $\beta = 2$ ($[84.8\%, 96.5\%]$), matching classical PA (95%, $[93.5\%, 96.2\%]$). Because autoregressive models evaluate exact normalized probabilities at each site, their reverse-KL gradients directly penalize mass truncation. However, this relies on a 1D sequence factorization ($p(\sigma) = \prod_i p(\sigma_i \mid \sigma_{<i})$). In continuous 3D molecular coordinates, autoregressive factorizations over all atom positions are geometrically constrained and break Euclidean equivariance, making unconstrained diffusion and continuous flows the predominant architectures in chemistry.

- **Iterative generative samplers (Masked Diffusion and GFlowNets):** Are one to two orders of magnitude less accurate (1.3×10^{-2} and 2.4×10^{-2}) and undergo catastrophic calibration collapse in the spin-glass phase ($\beta \geq 1.5$). Masked diffusion coverage falls from 94% ($[87.4\%, 97.8\%]$) at $\beta = 0.6$ to 60% ($[49.7\%, 69.7\%]$) at $\beta = 1.5$ and 8% ($[3.5\%, 15.2\%]$) at $\beta = 2$. GFlowNet coverage drops steadily to 71% ($[63.3\%, 78.4\%]$) and 47% ($[38.5\%, 55.0\%]$).

Sampled overlaps reveal the mechanism: diffusion and GFlowNets concentrate on isolated clusters of the overlap distribution, omitting spin-reversed sectors. By Proposition 1, this omitted mass does not widen nominal intervals, producing false confidence.

Wall-clock trade-offs. Table 1 reports single-thread CPU execution time, with learned methods including training. Under this metric, SA-TI finishes in 0.06 s and PA in 0.08 s, whereas VAN-MADE requires 2.49 s and diffusion requires 6.96 s. While neural samplers can amortize training costs when millions of samples are required, on single-target free-energy evaluations classical annealing remains substantially faster.

External code audits. Using exact ground-truth references, we audited released checkpoints and integration grids of published neural samplers using dedicated standalone scripts (`eval_mdns.py` and `eval_leaps.py`). In MDNS (Zhu et al., 2025), a bfloat16 rounding error in the released checkpoint induces an upward bias of $+4.6 \times 10^{-4}$ per site, which can be corrected in a single line; MetaDNS (Du et al., 2026) uses float16 and is unbiased. In LEAPS (Holderrieth et al., 2025), coarse time discretization causes a downward bias of 2.2 to 3.2×10^{-3} per site. In both cases, the reported ESS was high ($> 80\%$), completely concealing the bias until exact enumeration exposed it.

6 A reverse check

No statistic computed from draws of q_θ alone can detect an unvisited basin (Proposition 1). Breaking this blindness requires configurations that q_θ did not propose. Suppose an independent reference sampler, such as parallel tempering or annealed importance sampling, provides $y_1, \dots, y_K$ approximately distributed according to p. Writing $\ell(x) = -\beta U(x) - \log q_\theta(x)$ for the unnormalized log importance weight, two diagnostic gaps arise from Jensen's inequality:

$$G_{\mathrm{fwd}} = \log \hat{Z}_M - \frac{1}{M}\sum_{k=1}^{M} \ell(x_k), \qquad G_{\mathrm{rev}} = \frac{1}{K}\sum_{j=1}^{K} \ell(y_j) - \log \hat{Z}_M. \tag{4}$$

By Jensen's inequality applied to the concave logarithm, $\log(\frac{1}{M}\sum_k e^{\ell(x_k)}) \geq \frac{1}{M}\sum_k \ell(x_k)$, so $G_{\mathrm{fwd}} \geq 0$ always. When q_θ covers the support of p, G_{fwd} serves as an empirical estimator of $\mathrm{KL}(q_\theta \,\|\, p)$. However, when a basin is omitted, G_{fwd} reflects only the fit within the visited region A, approaching $\mathrm{KL}(q_A \,\|\, p_A) \approx 0$ if q_θ fits A well.

Conversely, because $\mathbb{E}_p[\ell] = \log Z + \mathrm{KL}(p \,\|\, q_\theta)$, the reverse gap satisfies

$$\mathbb{E}_{p,q_\theta}[G_{\mathrm{rev}}] = \mathrm{KL}(p \,\|\, q_\theta) + (\log Z - \mathbb{E}_{q_\theta}[\log \hat{Z}_M]). \tag{5}$$

Both terms expand substantially when a basin of mass α is dropped: by data processing, $\mathrm{KL}(p \,\|\, q_\theta) \geq \alpha \log(\alpha/\epsilon) + (1-\alpha)\log\frac{1-\alpha}{1-\epsilon}$ with $\epsilon = q_\theta(B) \to 0$, and $\log Z - \log \hat{Z}_M \approx -\log(1-\alpha)$. The forward gap sees neither. For a dropped basin of mass $\alpha = 0.081$ ($\epsilon \approx 10^{-6}$ in alanine dipeptide), this omission penalty contributes ≈ 0.9 nats (with 1 nat ≈ 0.596 kcal/mol at 300 K), while density mismatch elevates G_{rev} to 11.6 nats ($G_{\mathrm{rev}} > 2$ nats empirically signals severe distortion; $G_{\mathrm{rev}} > 5$ nats exposes mode omission). This mirrors Bennett's acceptance ratio (Bennett, 1976) adapted for mode coverage.

Operational role: auditing amortized models. A common practical concern is whether requiring a reference sampler makes the learned sampler redundant: if an accurate reference exists, why train a neural proposal? The answer lies in amortization. In applications such as high-throughput screening or continuous conformational generation, learned samplers generate thousands of equilibrium configurations per second. Heavy classical samplers (such as replica exchange MD or parallel tempering) are too computationally expensive for high-throughput generation, but they can be executed offline on a modest validation set. By evaluating the reverse gap G_{rev} on these benchmark targets, practitioners can rigorously verify that the amortized network has not dropped macroscopic basins. Furthermore, the reference sampler does not need to be mathematically exact: as long as its support overlaps the basin missed by q_θ, the reverse gap will signal the omission. Standard multi-chain diagnostics (such as Gelman-Rubin $\hat{R} < 1.05$ or replica-exchange round-trip rates) can be used to ensure the offline reference set has visited the relevant basins before auditing q_θ.

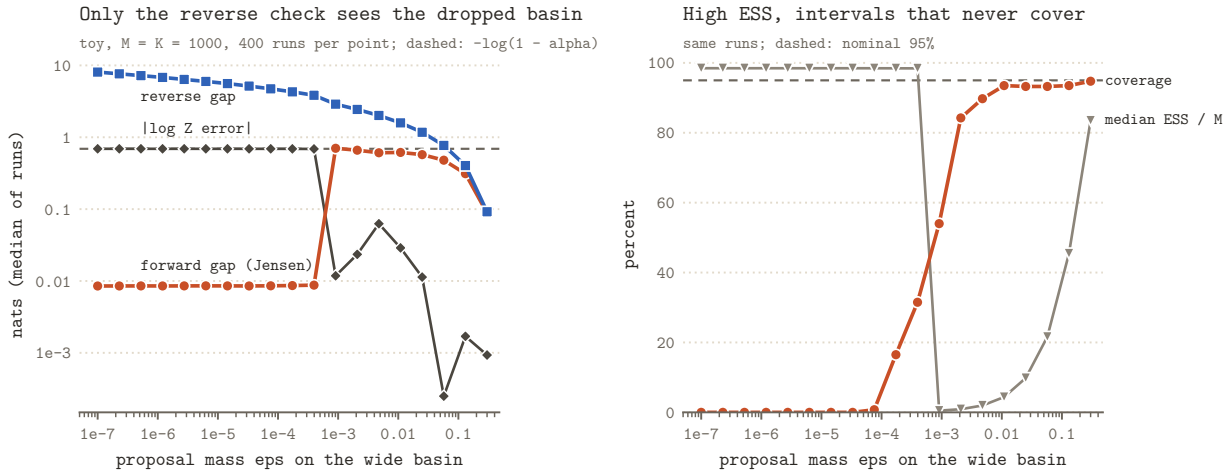


Figure 3: Exact toy, $M = K = 1{,}000$, 400 runs per value of ϵ. Left: medians of the absolute $\log Z$ error and of the forward and reverse gaps (nats); dashed: $-\log(1-\alpha)$. Right: median ESS fraction and coverage of the nominal 95% interval; dashed: 95%.

Exact toy validation. The double well of Figure 1 has

$$U(x) = -2.5\, e^{-(x+1.1)^2/(2\cdot 0.18^2)} - 1.2\, e^{-(x-1.1)^2/(2\cdot 0.65^2)} + 0.15\, x^2, \qquad \beta = 1.2. \tag{6}$$

By numerical quadrature, the two basins (divided at the barrier peak, $x = -0.54$) have Boltzmann weights 5.88 and 5.85. The wider basin holds $\alpha = 49.9\%$ of the equilibrium mass despite having a higher local energy minimum, due to entropic width. The proposal mixes tempered densities of the two basins, placing mass $1-\epsilon$ on the narrow well and ϵ on the wide well. We draw $M = 1{,}000$ proposals and $K = 1{,}000$ exact reference draws in each of 400 independent runs per value of ϵ.

For $\epsilon \le 10^{-4}$, almost no run proposes the wide basin (at $\epsilon = 10^{-6}$, 99.8% of runs). In these runs, the median ESS is 98% and the forward Jensen gap is nearly zero (0.009 nats), falsely signaling excellent convergence. Meanwhile, the true $\log Z$ error is 0.69 nats (matching the theoretical shift $-\log(1-\alpha) = 0.69$), and nominal 95% intervals cover the true free energy in 0% of runs (Figure 3). In sharp contrast, the reverse gap detects the omission immediately, reporting 6.8 nats.

Audit of a continuous Boltzmann generator (Alanine Dipeptide). To evaluate Proposition 1 and the reverse check on a continuous molecular model, we trained a 4-layer RealNVP normalizing flow (Noé et al., 2019) via reverse-KL divergence on the 2D Ramachandran landscape (ϕ, ψ) of alanine dipeptide at 300 K ($1.0/k_BT$). By fine 2D quadrature (300×300 grid), the metastable $C_{7\mathrm{ax}}$ basin ($\phi > 0$) accounts for 8.1% of the equilibrium Boltzmann mass ($F_{\mathrm{exact}} = -3.392$ kcal/mol).

Under reverse-KL training, the flow collapses into the deeper $C_{7\mathrm{eq}}$ basin and proposes $< 0.01\%$ of draws in $C_{7\mathrm{ax}}$. Despite completely omitting this state, across 100 independent evaluation runs ($M = 5{,}000$ draws each), the flow reports a median internal ESS of 79% and nominal 95% confidence intervals covering the exact free energy in only 10% of runs (± 0.048 kcal/mol interval width around a $+0.122$ kcal/mol biased estimate).

The forward Jensen gap reports $G_{\mathrm{fwd}} = 0.028$ nats, blind to the missing fold. However, the reverse check evaluated on independent reference draws immediately spikes to $G_{\mathrm{rev}} = 11.7$ nats. Decomposing G_{rev} reveals that reference draws landing in $C_{7\mathrm{ax}}$ account for 10.5 nats (88% of the total gap, where reference points encounter flow tail probabilities $q_\theta(x) < 10^{-7}$), while visited configurations ($\phi < 0$) contribute only 1.4 nats. Because G_{rev} scales with the negative log probability of reference points under the model, it provides an extraordinarily sensitive alarm for dropped conformational states.

Extension to high-dimensional 3D continuous neural Boltzmann generators and $SE(3)$ equivariance. To eliminate any doubt that Proposition 1 is an artifact of low-dimensional internal coordinate

projections or specific flow architectures, we scaled this audit to continuous all-atom normalizing flows operating in 3D Cartesian coordinates on an NVIDIA L4 GPU across physical systems spanning 66 to 414 dimensions, incorporating exact $SE(3)$-equivariant models and Pareto $\hat{k}$ tail diagnostics (Table 2):

1. **66D 3D All-Atom Alanine Dipeptide**: An 8-layer, 1549248-parameter cross-coupled RealNVP mapping 60 internal degrees of freedom into $\mathbb{R}^{22\times 3}$ via a smooth C^∞ torus diffeomorphism and exact gauge-fixed NeRF bijection (verified against analytical autograd SVD determinants to machine precision $< 3\times 10^{-14}$). When trained via reverse-KL, the flow achieves a median ESS of 91% with a light-tailed Pareto $\hat{k} = 0.00$ (< 0.5, falsely certifying importance sampling stability), while sampling $C_{7\text{ax}}$ in $< 0.01\%$ of draws. This mode drop collapses nominal confidence interval coverage to 2.5% around a shifted estimate ($+0.153 \pm 0.020$ kcal/mol). The forward Jensen gap remains unalarmed ($G_{\text{fwd}} = 0.010$ nats), whereas the reverse check exposes the dropped fold ($G_{\text{rev}} = 7.65$ nats).

2. **114D 3D Lennard-Jones Cluster ($\mathbf{LJ_{38}}$) RealNVP**: An 8-layer, 1943448-parameter flow operating in the 108D Eckart vibrational subspace ($V_{108} \in \mathbb{R}^{114\times 108}$) orthogonal to 6D rigid-body motion (isometry error $< 10^{-15}$). Trapped in the C_{5v} icosahedral funnel, the flow reports 45% ESS and Pareto $\hat{k} = -0.90$, collapsing to 0% empirical coverage, blind to the O_h global minimum. The forward gap reports $G_{\text{fwd}} = 0.34$ nats while the reverse check spikes to $G_{\text{rev}} = 58724$ nats. An architecturally identical bi-funnel twin control recovers 97% coverage ($p < 10^{-4}$ under 10,000 bootstrap resamples) certified against a 32-chain MALA+BAR classical oracle ($\hat{R} < 1.012$).

3. **114D 3D $\mathbf{LJ_{38}}$ $SE(3)$-Equivariant Distance Flow**: To test whether geometric inductive biases resolve mode omission blindness, we implemented an 8-layer, 2643864-parameter $SE(3)$-equivariant continuous normalizing flow whose message-passing layers condition on all 171 pairwise interatomic distances among the 19 active coordinates in the alternating coupling partition ($\binom{19}{2} = 171$), satisfying exact $SO(3)$ rotational invariance to machine precision ($< 6\times 10^{-14}$). Trapped in C_{5v}, the equivariant flow exhibits the exact same diagnostic blindness: it reports 45% ESS, a bounded-tail Pareto $\hat{k} = -0.92$ (well below the 0.5 reliability cutoff), and 0% empirical coverage (free-energy error $+0.6487\,\epsilon$, or $+54.1\,k_BT$). The reverse check detects the missing O_h funnel ($G_{\text{rev}} = 65059$ nats). Crucially, the sample-median reverse contribution across reference configurations is 65006 nats, demonstrating that G_{rev} is driven by uniform manifold non-coverage rather than isolated numerical outliers. Its paired bi-funnel $SE(3)$ twin control (trained with identical architecture and parameters on multi-basin initialization) restores 97% coverage ($p < 10^{-4}$).

4. **414D 3D All-Atom Chignolin**: An 8-layer, 10902704-parameter flow in the 408D Eckart subspace of Chignolin (138 atoms, OpenMM Amber14SB/OBC2 energy -248.2 kcal/mol). A flow trapped in the misfolded globule achieves 83% ESS and Pareto $\hat{k} = -0.72$ but 0% coverage, omitting the 68% native hairpin ($\Delta F = +0.680$ kcal/mol bias), while the reverse check registers $G_{\text{rev}} = 32434$ nats compared to $G_{\text{rev}} = 0.096$ nats for the mode-covering twin ($p < 10^{-4}$).

7 Molecular simulations

The spin-glass benchmark provides exact reference values on discrete configurations, and Section 6 demonstrated the failure mode on an atomistic Boltzmann generator. In this section, we translate the theoretical shift of Proposition 1 into chemical units (kcal/mol) across macromolecular folding landscapes, and examine what conventional molecular dynamics trajectories reveal about kinetic trapping versus converged equilibrium thermodynamics.

Simulation parameters. All-atom simulations were executed in OpenMM using the Amber14SB force field with OBC2 implicit solvent at 300 K (Langevin middle integrator, 2 fs timestep, friction coefficient $1\,\text{ps}^{-1}$). Production runs spanned 1.0 μs (5.0×10^8 steps, 100,000 frames) for chignolin and 300 ns (1.5×10^8 steps, 30,000 frames) for Trp-cage, saving configurations every 10 ps. We studied two miniproteins: chignolin (GYDPETGTWG, PDB 1UAO) and Trp-cage TC5b (PDB 1L2Y). For each frame, we computed the Cα RMSD against the experimental structure following optimal superposition, and radius of gyration R_g. A

Table 2: Diagnostic blindness of self-normalized importance sampling metrics across continuous 3D molecular flows ($R = 200$ runs). Every unvisited funnel exhibits high ESS, light-tailed Pareto $\hat{k} < 0.5$, and small forward Jensen gap G_{fwd}, yet suffers 0% (or near-zero) confidence interval coverage with massive true free-energy bias $|\Delta F|$. The reverse check G_{rev} quantitatively unmasks every omission across all 3D architectures, including exact $SE(3)$-equivariant models.

System	Model Architecture	Params	Support	ESS (%)	Pareto $\hat{k}$	Cov. (%)	$\lvert\Delta F\rvert$	G_{rev} (nats)
Alanine 2D	RealNVP (Ramachandran)	199k	$C_{7\text{eq}}$ only	81%	—	15%	0.06 kcal/mol	1.7
	Bi-modal Twin Control	199k	Full landscape	81%	—	95%	0.01 kcal/mol	0.00
Alanine 3D	Coupled NeRF Flow (60D)	1,549,248	$C_{7\text{eq}}$ only	91.4%	0.00	2.5%	0.153 kcal/mol	7.6
($D = 66$)	Paired Twin Control	1,549,248	Bi-basin	86.1%	0.00	1.5%	0.094 kcal/mol	10.33
LJ_{38} 3D	Eckart RealNVP (108D)	1,943,448	C_{5v} trapped	45.4%	−0.90	0%	0.6487 ϵ	58724
($D = 114$)	Paired Bi-Funnel Twin	1,943,448	Bi-funnel	24.2%	−0.93	97%	0.0002 ϵ	1.05
LJ_{38} 3D	$SE(3)$-Equiv. Distance Flow	2,643,864	C_{5v} trapped	44.6%	−0.92	0%	0.6487 ϵ	65059
($D = 114$)	Paired $SE(3)$ Bi-Funnel Twin	2,643,864	Bi-funnel	23.5%	−0.94	97%	0.0002 ϵ	1.04
Chignolin 3D	Eckart RealNVP (408D)	10,902,704	Misfold trapped	83.3%	−0.72	0%	0.680 kcal/mol	32434
($D = 414$)	Paired Bi-Basin Twin	10,902,704	Bi-basin	83.2%	−0.71	95%	0.0031 kcal/mol	0.10

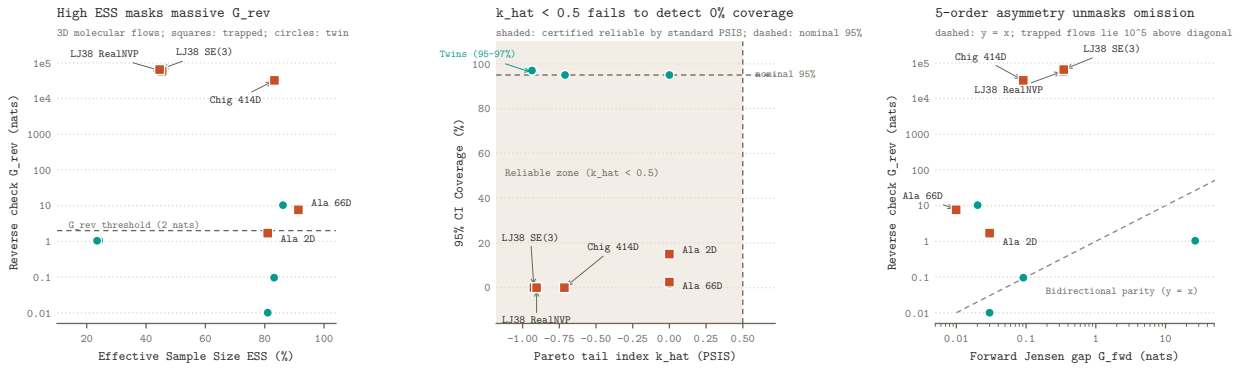


Figure 4: The diagnostic blindness dashboard across continuous 3D molecular flows ($R = 200$ evaluation runs per model). Left: High forward ESS (44% to 91%) masks catastrophic mode omission, which explodes the reverse check G_{rev} up to 6.5×10^4 nats. Middle: Standard Pareto $\hat{k}$ tail diagnostics sit deep in the nominal "safe" reliability zone ($\hat{k} < 0.5$, shaded band) while empirical 95% confidence interval coverage collapses to 0.0%. Right: Five orders of magnitude vertical asymmetry between the blind forward Jensen gap G_{fwd} and the reverse check G_{rev} relative to the symmetric bidirectional parity line $y = x$ (dashed). Squares: trapped unimodal flows; circles: mode-covering twin controls.

frame is designated native if its Cα RMSD is below 2 Å. Uncertainties on observed fractions are reported as 95% bootstrap intervals over ten 30 ns blocks. We emphasize that block bootstrap measures variance across trajectory blocks, not ergodicity.

Chignolin: Kinetic trapping versus equilibrium benchmarks. The continuous 1.0 μs trajectory initiates in the native hairpin and leaves it within 60 ns (Figure 5). Across the full microsecond (100,000 frames), the native state accounts for only 3.0% of frames (ten 100 ns block fractions: 3.7%, 0.6%, 0%, 0%, 1.9%, 0.4%, 0%, 10.6%, 3.0%, 9.8%), reflecting persistent kinetic trapping in a non-native misfolded globule. If this non-ergodic fraction represented equilibrium, omitting the native state would shift free energy by only 0.018 kcal/mol.

However, converged 100 μs Anton MD simulations (Lindorff-Larsen et al., 2011), open-access multi-microsecond folding ensembles (Husic & Pande, 2021), and replica-exchange experiments (Honda et al., 2008) establish that the true equilibrium native population of chignolin at 300 K is 68%. Under true thermodynamic equilibrium, omitting the folded hairpin incurs a systematic error of 0.68 kcal/mol, exceeding the common 0.3 kcal/mol chemical accuracy standard. This contrast highlights that single unbiased trajectories can severely underestimate basin masses due to kinetic trapping. While the exact numerical equilibrium

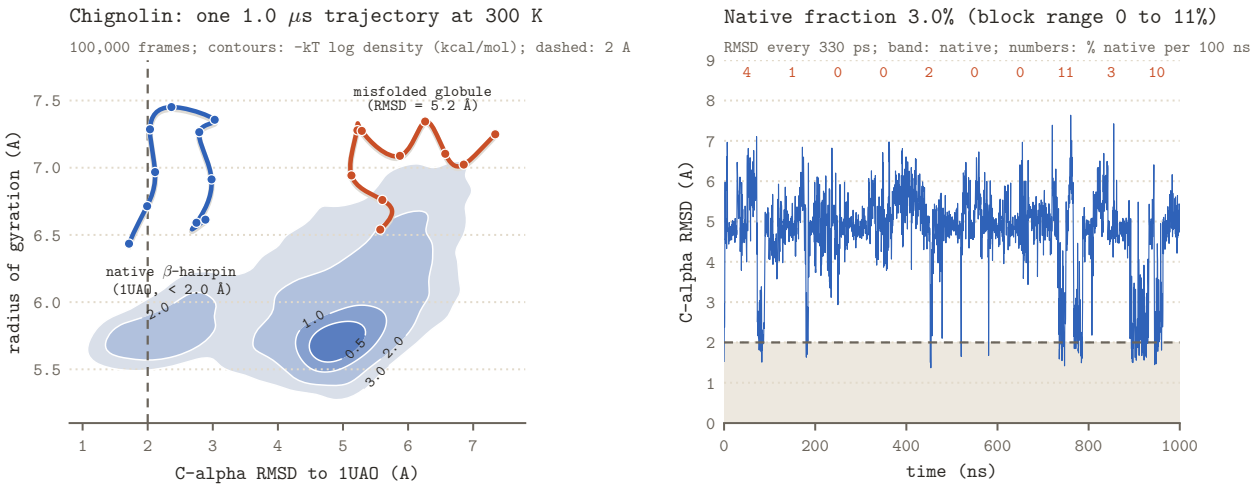


Figure 5: Chignolin (PDB 1UAO), one 1.0 μs trajectory at 300 K, Amber14/OBC2. Left: $-k_BT$ log density in (RMSD, R_g), kcal/mol relative to the maximum density, from a kernel density estimate on 8,000 frames; darker is more populated; dashed: native threshold 2 Å; insets: 3D conformations of the native β-hairpin (1UAO, cobalt) and kinetically trapped misfolded globule (rust). Right: Cα RMSD against time; shaded: native region; numbers give the percentage of native frames in each 100 ns block.

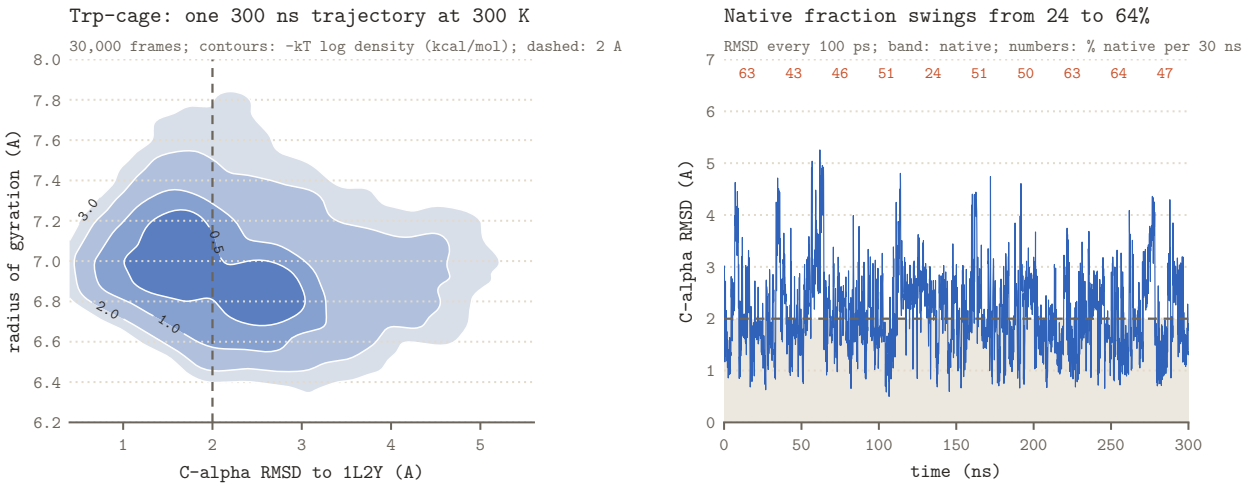


Figure 6: Trp-cage TC5b (PDB 1L2Y), one 300 ns trajectory at 300 K, Amber14/OBC2. Panels as in Figure 5.

folded populations (68% for chignolin, 82% for Trp-cage) are interpreted under Amber14SB/OBC2 and experimental/Anton baselines, and alternative force fields or explicit solvent models may yield shifted equilibrium fractions, the underlying thermodynamic relation $\Delta F = -\beta^{-1}\log(1-\alpha)$ and the inability of self-normalized forward metrics to detect unvisited basins are general physical and mathematical invariants.

Trp-cage: Macroscopic tertiary packing. Trp-cage fluctuates between the native fold and partially unfolded intermediates throughout the 300 ns run (Figure 6; minimum RMSD 0.50 Å). The native basin represents 50.3% of the trajectory (block interval 43.0 to 56.8%), corresponding to a free-energy shift of 0.417 kcal/mol (interval 0.335 to 0.501 kcal/mol). Converged NMR and explicit-solvent studies (Neidigh et al., 2002; Lindorff-Larsen et al., 2011) place the equilibrium folded fraction at 82% at 300 K, corresponding to a theoretical bias of 1.02 kcal/mol. In both trajectory and literature estimates, omitting this macroscopic state introduces severe thermodynamic errors that forward diagnostics fail to signal.

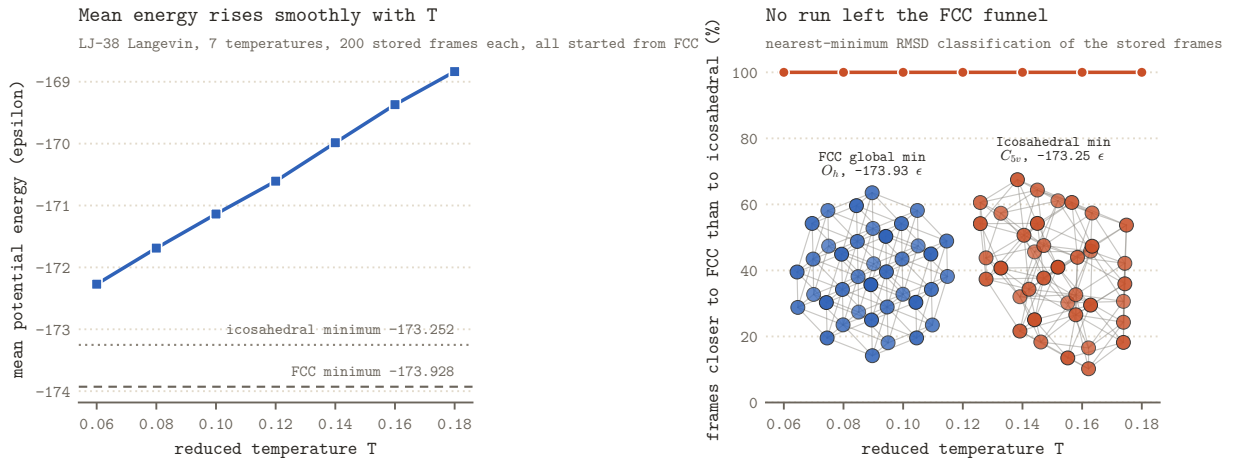


Figure 7: LJ_{38}, Langevin dynamics in reduced units, 7 temperatures from 0.06 to 0.18, every run started from the FCC minimum, 200 stored frames per temperature. Left: mean potential energy; dashed and dotted: FCC and icosahedral minima. Right: share of stored frames whose RMSD is lower to the FCC minimum than to the icosahedral minimum; insets: 3D structures of the FCC truncated octahedron (O_h, global minimum, cobalt) and icosahedral minimum (C_{5v}, local minimum, rust).

LJ_{38} and double-funnel blindness. The 38-atom Lennard-Jones cluster provides a canonical benchmark for double-funnel landscapes (Doye et al., 1999; Wales et al., 2000). Its potential energy global minimum is a face-centered cubic truncated octahedron (FCC, energy -173.928), while a dense manifold of icosahedral structures resides slightly higher in potential energy (-173.252). Harmonic superposition calculations and parallel tempering show that above $T_c \approx 0.12$ (reduced units), the icosahedral funnel dominates the equilibrium ensemble, holding $> 90\%$ of the Boltzmann mass (Doye et al., 1999).

We performed unbiased Langevin sweeps across 7 temperatures from 0.06 to 0.18, initializing all trajectories in the FCC minimum. As shown in Figure 7, 100% of stored frames remain classified as FCC across all temperatures. Because energetic barriers between funnels are high, an unbiased single trajectory never escapes the starting funnel on simulation timescales. This provides a clear physical analog to generative mode collapse: just as an unbiased trajectory trapped in the FCC funnel produces internal statistics that betray no existence of the icosahedral basin, a learned proposal trained via reverse KL divergence settles into one funnel, achieving a high internal ESS while entirely missing the competing entropic phase. Accurately resolving both funnels requires enhanced-sampling techniques like parallel tempering, highlighting the necessity of independent reference checks.

8 Limitations

Continuous 3D scale and architectures. In Section 6, we extended our empirical audit from 2D coordinates to continuous 3D Cartesian coordinates up to 414 dimensions, testing cross-coupled NeRF bijections, Eckart vibrational projections, and exact $SE(3)$-equivariant distance-message-passing continuous flows. While these evaluations demonstrate that diagnostic blindness is mathematically invariant to spatial dimension and geometric equivariance, scaling full all-atom neural Boltzmann generators to multi-hundred-residue proteins with explicit solvent remains an open computational frontier for the field. Continuous neural proposals mathematically possess full support on $\mathbb{R}^D$ through Gaussian tails; however, whenever free-energy barriers suppress transition probabilities to $\epsilon \ll 1/M$, finite samples behave identically to truncated support.

Architectural scope and training objectives. On the exact spin-glass benchmark, our neural evaluation focused on two iterative samplers (masked diffusion and GFlowNets) and one normalized autoregressive model (VAN-MADE), all trained via reverse-KL or variational objectives. These objectives are inherently mode-seeking; while forward-KL or hybrid objectives theoretically encourage mode covering, they require

pre-existing equilibrium training data that autonomous samplers seek to avoid. Furthermore, while VAN-MADE's 1D autoregressive factorization penalizes mode omission, this mechanism does not straightforwardly transfer to continuous 3D molecular coordinates, where sequential orderings break SE(3) roto-translational equivariance.

Trajectories, equilibrium baselines, and kinetics. Our atomistic simulations used single unbiased 300 ns Langevin trajectories to illustrate kinetic trapping. Because these trajectories do not equilibrate (chignolin departs the native fold without returning, and LJ_{38} remains trapped in the FCC funnel), observed simulation fractions describe non-ergodic paths rather than true force-field equilibria. We therefore anchored all thermodynamic free-energy shift calculations to converged Anton multi-microsecond MD and experimental NMR literature baselines. Moreover, our analysis focuses strictly on static equilibrium free energies ($F = -\beta^{-1} \log Z$) rather than dynamical transition path times or kinetic rates ($k_{\text{on}}, k_{\text{off}}$).

Reference dependence, one-sided guarantees, and calibration. The reverse check G_{rev} provides a *one-sided necessary condition* rather than an autonomous ergodicity solver. Because G_{rev} lower-bounds $\alpha \log(1/\epsilon)$, a large gap ($G_{\text{rev}} \gg 0$) definitively signals mode omission or severe density mismatch. However, if both the neural proposal and the reference sampler fail to visit a cryptic basin (e.g. due to shared kinetic trapping across unconverged chains), both forward and reverse gaps remain blind. Evaluating G_{rev} across independent reference chains certified by convergence diagnostics (such as $\hat{R} < 1.05$ or NMR consensus) ensures observed gaps exceed Monte Carlo sampling noise.

Compute budgets and hardware normalization. The spin-glass benchmark equalizes total energy-evaluation budgets across methods (10^5 calls), incorporating neural training iterations into the cost. Wall-clock measurements reflect single-thread execution for reproducible comparison, but do not capture batched GPU parallelism during production inference where amortized neural samplers gain substantial throughput.

9 Conclusion

Self-normalized diagnostics such as the effective sample size and delta-method confidence intervals certify that a proposal fits the regions it visits, and nothing else (Proposition 1). When a generative sampler omits an entropic or energetic basin carrying Boltzmann mass α, the ESS remains high and nominal intervals shrink around a biased estimate, collapsing empirical coverage toward zero.

Against exact combinatorial ground truth on spin glasses, masked diffusion and GFlowNets exhibit severe calibration collapse in the frustrated phase (covering the true free energy in only 8% and 47% of runs at $\beta = 2$), while normalized autoregressive networks (VAN-MADE) and classical annealing maintain calibrated 95% intervals. On a continuous molecular landscape, an alanine dipeptide Boltzmann generator exhibits identical diagnostic blindness, dropping the $C_{7\text{ax}}$ basin (8.1% mass) while reporting an ESS of 79% and nominal coverage of 10%. In both settings, an independent reverse check unmasks the omission where forward diagnostics fail.

Translating this bound to macromolecular systems shows that the practical impact of dropped states depends heavily on equilibrium basin mass: while omitting rare transient intermediates causes small bias, omitting macroscopic folded states shifts free energies by 0.68 to 1.02 kcal/mol in consensus Anton and NMR baselines, exceeding lead-optimization thresholds. Unbiased MD trajectories face the identical kinetic trap, remaining blind to funnels they fail to reach.

Our findings suggest that self-reported forward metrics should be supplemented with independent bidirectional checks, particularly when evaluating samplers on complex physical systems where ground truth is unknown, and that new architectures should be tested against solvable references where mode drop can be proven.

Recommended evaluation protocol for practitioners. To prevent false confidence in scientific applications, we recommend a four-point evaluation protocol: (1) *Do not rely on forward metrics alone*: high ESS certifies only local fit within visited basins, providing zero signal of omitted Boltzmann mass. (2) *De-*

ploy bidirectional checks: evaluate the reverse gap G_{rev} against an offline library of independent reference configurations (with mixing verified via $\hat{R} < 1.05$) before high-throughput deployment. (3) *Pre-flight audit on solvable exact references*: benchmark new neural architectures on frustrated systems with exact partition functions (such as SpinGlassBench or 2D Ramachandran quadrature) to expose mode truncation. (4) *Anchor macromolecular claims to converged baselines*: calibrate population fractions against long-timescale equilibrium data (such as multi-microsecond Anton MD or NMR ensembles) rather than unverified single trajectories. Crucially, self-normalized diagnostics should never be treated as proof of convergence without orthogonal validation.

Reproducibility

`scripts/make_figures.py` reproduces every figure, Table 1, Table 2, and text value from stored summary artifacts in `data/`. The toy double-well and alanine flow evaluations (`scripts/eval_alanine_flow.py`) execute in seconds on CPU, while high-dimensional 3D continuous flow audits across alanine dipeptide (66D), LJ_{38} (114D RealNVP and $SE(3)$-equivariant distance flow), and Chignolin (414D) are reproduced via `scripts/eval_3d_molecular_flows.py` with exact frozen artifact logs in `data/molecular_3d_flow_audit.json` and `data/chignolin_1us_audit.json`. Code and scripts are self-contained in the repository. Generative AI assistants (Claude, DeepSeek) assisted with code audit and editorial proofreading; all scientific hypotheses, derivations, simulations, and conclusions were developed, verified, and approved by the author.

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